# Free-body diagrams

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Draw only the external forces on one chosen object.

**SUCCESS CRITERIA - I can...**

1. Choose one object and list its external interactions.
2. Draw force arrows in physically correct directions.
3. Remove velocity and net force as extra interaction arrows.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “normal force” and “tension” to explain your model or result. Sentence starters are optional.

**normal force**: a support force perpendicular to a contact surface

**tension**: the pulling force through a stretched rope or cable

**weight**: the gravitational force on an object

## 01 Warm-up: bring back an earlier idea

An a-versus-$F_{net}$ graph has slope 1.5 $(m/s^2)/N$.
**Find** mass and acceleration at 2.5 N.
**Explain** why a driving-force plot can have an offset.

**Word help:** mass = a measure of how strongly an object resists acceleration

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $\Sigma F_x = ma_x$; $\Sigma F_y = ma_y$; $F_g = mg$.
Draw only forces on the chosen object. x and y name directions. Use g = 9.8 N/kg unless another value is given.

**Worked reasoning:** A lamp hangs motionless from one vertical cable. With weight 12 N down, the cable tension is 12 N up. Only forces acting on the lamp belong in its diagram.

**Your turn: Draw** the lamp as a labeled point with its two force arrows.
**Explain** why its force on the cable is not another arrow on this diagram.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .



Add and label force arrows.

Lamp: draw the two 12 N forces acting on it.

**Marzano check - circle:** ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 03 Motionless lamp

A 2 kg lamp hangs at rest from one vertical rope.
**Use** g = 9.8 N/kg.
**Draw** and label the forces on the lamp, including their magnitudes.

**Word help:** at rest = not moving in the chosen reference frame

**Sentence starter:** The … in my model represents … . Its label shows … .

Add and label force arrows.

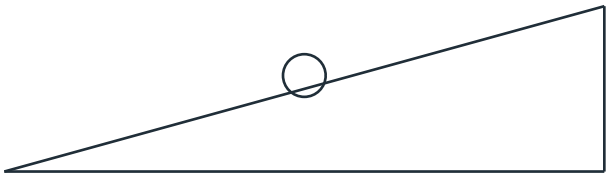
The dot represents the object. Object: 2 kg lamp

**Marzano check - circle:** 1 2 3

## 04 A rough ramp

A block rests on a rough incline. Add weight, normal force and static friction to the ramp model.
**Explain** why weight points vertically downward rather than perpendicular to the surface.

**Sentence starter:** Under the assumption …, the evidence … supports … because … .



**Marzano check - circle:** 1 2 3

## 05 Critique the extra arrows

A student's free-body diagram includes weight, normal force, friction, a "velocity force," and an extra "net force." Redraw a correct diagram for a box sliding right on a rough level floor after release.
**Explain** each removal.

**Sentence starter:** Under the assumption …, the evidence … supports … because … .

Add and label force arrows.

Box after release; it slides right.

**Marzano check - circle:** 1 2 3

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Continue down each column.

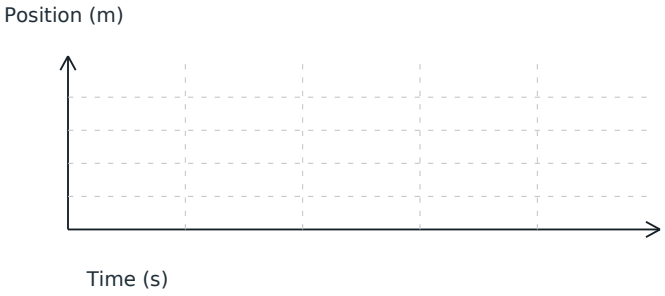
Sentence starters are optional. You may plan in any language.

## 06 Review: connect an earlier idea

A straight position-time segment goes from (2 s, -4 m) to (6 s, 20 m).
**Find** velocity, interpret its units and predict x at 8 s if motion continues.
**Label** the model with the given positions and any prediction.

**Word help:** velocity = rate of position change, with direction

**Sentence starter:** I predict … . My model supports this because … .




**Marzano check - circle:** 1 2 3

## 07 Exit check

A box slides to the left on a rough level floor after release.
**Draw** all forces; state what can be concluded about vertical and horizontal acceleration.

**Sentence starter:** The … in my model represents … . Its label shows … .

Add and label force arrows.

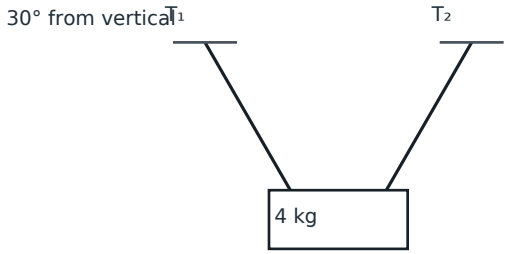
Box slides left after release on a rough level floor.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

Two identical cables support a stationary 4 kg object. Each cable is 30° from vertical.
**Use** g = 10 N/kg. Add force arrows to the model and calculate the tension in each cable.

**Sentence starter:** I found … [unit]. This value means … in this situation.




**Marzano check - circle:** 1 2 3